

Chapter 05

Localization

Pervasive Computing

MOBC 3302

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Content

- Pervasive system usually provides its services according to the current location/position of the user. So, in this chapter, we will briefly talk about Localization.



Content

- Location-Based Service
- Localization
- Localization Tasks
- Types of Location
- Types of Localization
- Localization Techniques



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- **Location-Based Service**
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Location-Based Service

- In a lot of services, the pervasive system depends/uses the **current location** of the user to correctly serve him.
- For example:
 - RFID
 - Self-driving car
 - Smart home (in some services)
 - Navigation systems
 - Traffic Management System
 - Burglar detection system

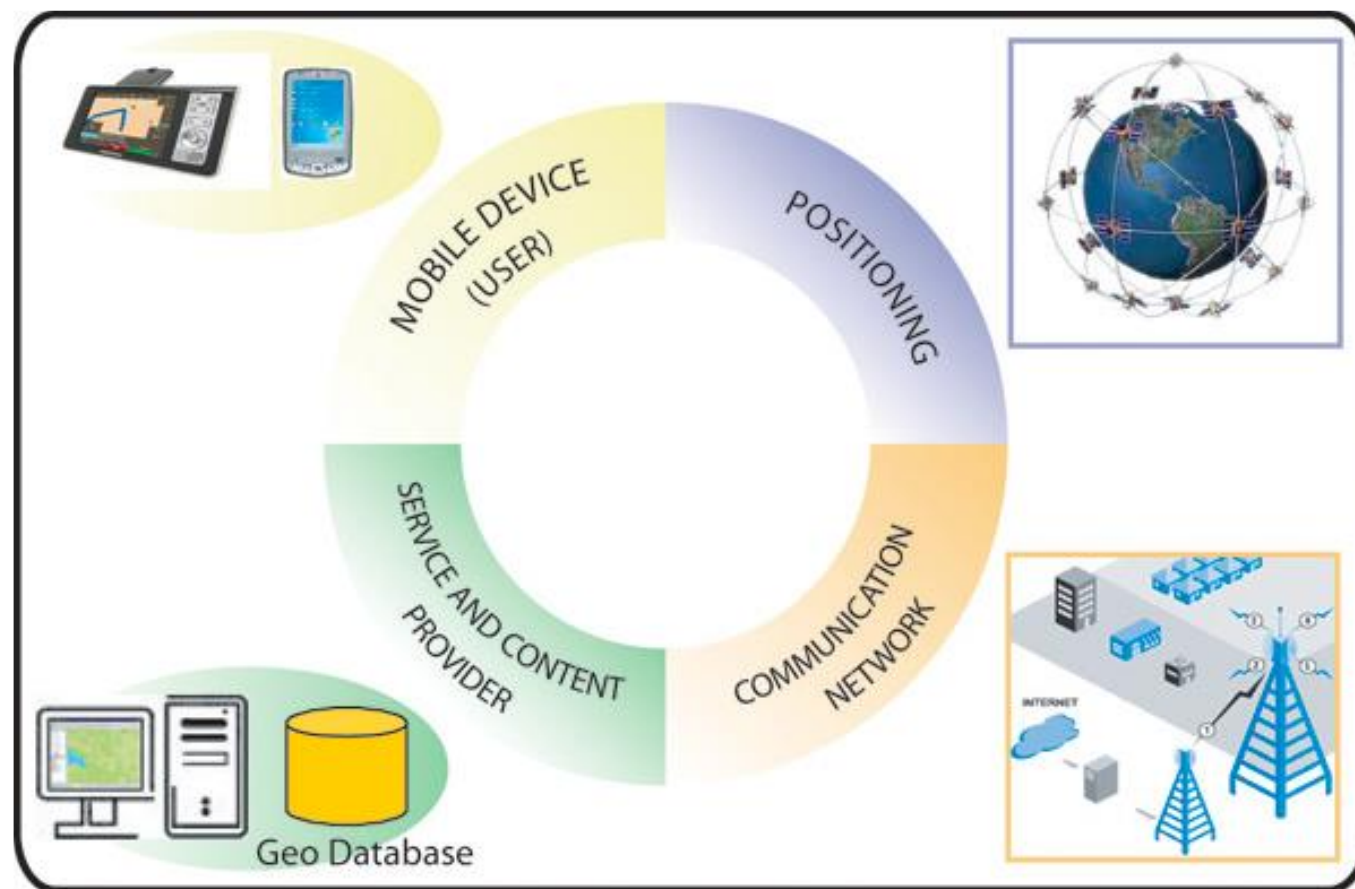


Location-Based Service

- A **location-based service (LBS)** is a software service for portable applications that requires knowledge about **where the device** is geographically **located**.
- The application collects **geodata**, which is data gathered in **real time** using one or more location tracking technologies.
- Location-based services integrate data from various resources, including Global Positioning System (GPS) satellites, cellular tower pings and short-range positioning beacons, to provide services based on the user's geographical location.



Location-Based Service





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Localization

- In the software, **Localization** is the process of **real-time** or non-real-time specifying the **current position** of a device, in a limited place or globe.
- For example:
 - GPS to determine the current coordinates of mobile in the map
 - Determine the current position of a file/device/hardware within a building



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Localization Tasks

- Actually, Localization can be used for tasks other than positioning.
- Like:
 - Positioning
 - Determine the current position on a map or a place
 - Addressing
 - Determine a readable address (city, street, etc)
 - Geocoding
 - Convert readable address to coordinates
 - Routing
 - Directing a user to walk or drive on some roads, to reach a destination
 - Calculating distance
 - Compute the distance between two locations
 - Step counter
 - Count steps in which user walks it



Localization Tasks



Action	Questions	Operations
orientation & localisation locating	where am I? where is {person object}?	positioning, geocoding, geodecoding
navigation navigating through space, planning a route	how do I get to {place name address xy}?	positioning, geocoding, geodecoding routing
search searching for people and objects	where is the {nearest most relevant &}{person object}?	positioning, geocoding, calculating distance and area, finding relationships
identification identifying and recognising persons or objects	{what who how much} is {here there}?	directory, selection, thematic/ spatial, search
event check checking for events; determining the state of objects	what happens {here there}?	



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Types of Location

- There are three types of locations:
 - Absolute
 - Relative
 - Symbolic



Types of Location (Absolute)

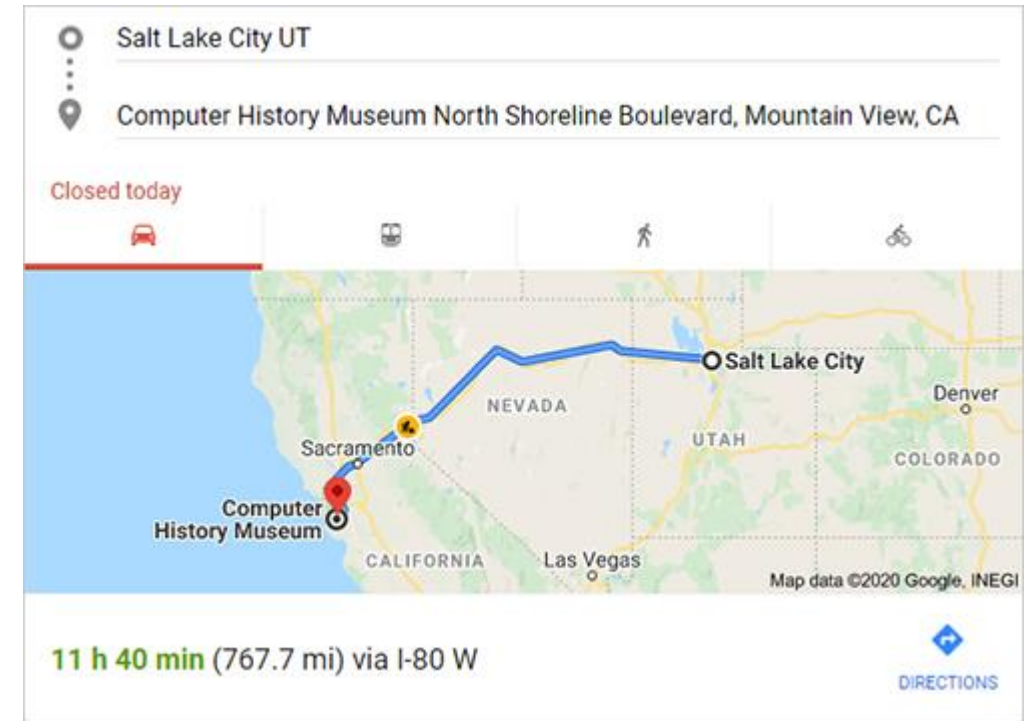
- **Absolute Location**, known as a geometrical location, which is presented **on the map** by latitude and longitude. The latitudinal location of any place or area is of much significance.





Types of Location (Relative)

- **Relative location** also called as the geographical situation. It is the location of a point or place in **relation to another** point or place.





Types of Location (Symbolic)

- **Symbolic location** is a location representation based on the descriptive of **the location itself**, such as a city or a named room. It is also known as descriptive, hierarchical or topological location.
- For example:
 - Room P201
 - Next door of the chicken



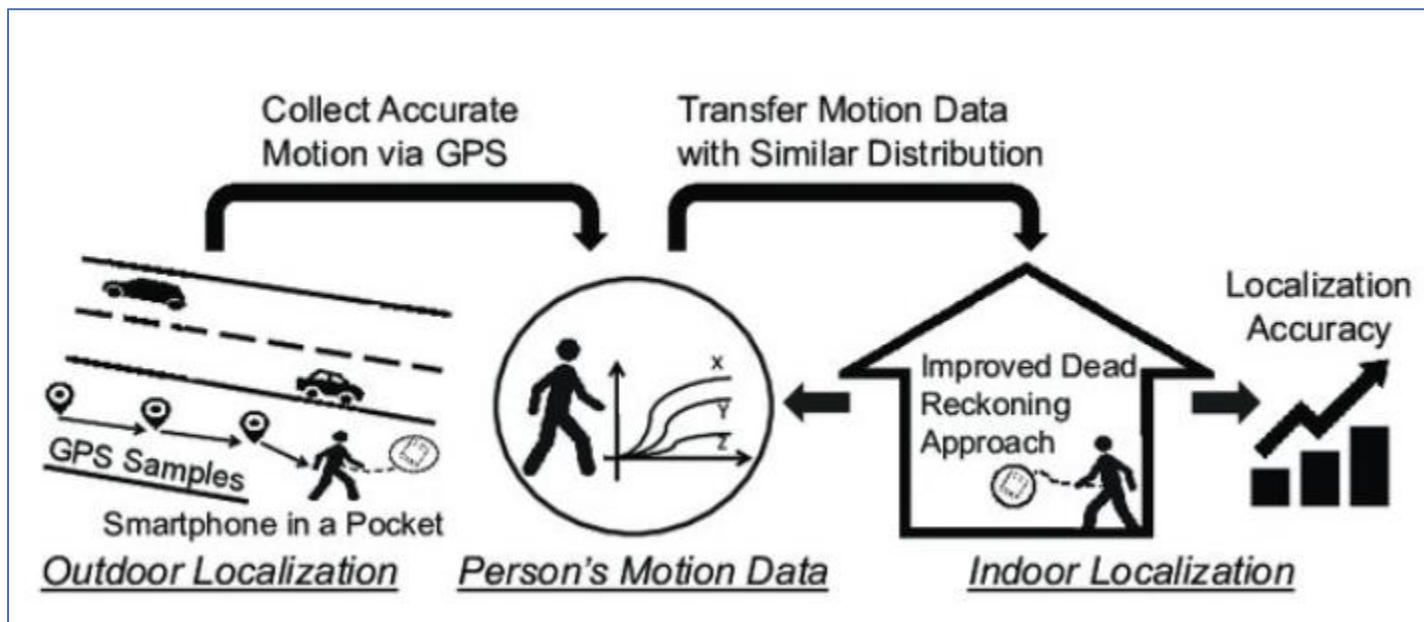
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Types of Localization

- There are two types of localization:
 - Outdoor Localization
 - Indoor Localization





Types of Localization (Outdoor Localization)

- **Outdoor Localization** refers to make localization on **non-limited-geographically places**, like a map or an open city.





Types of Localization (Indoor Localization)

- Indoor Localization refers to make localization on **limited-geographically places**, like a building or factory.

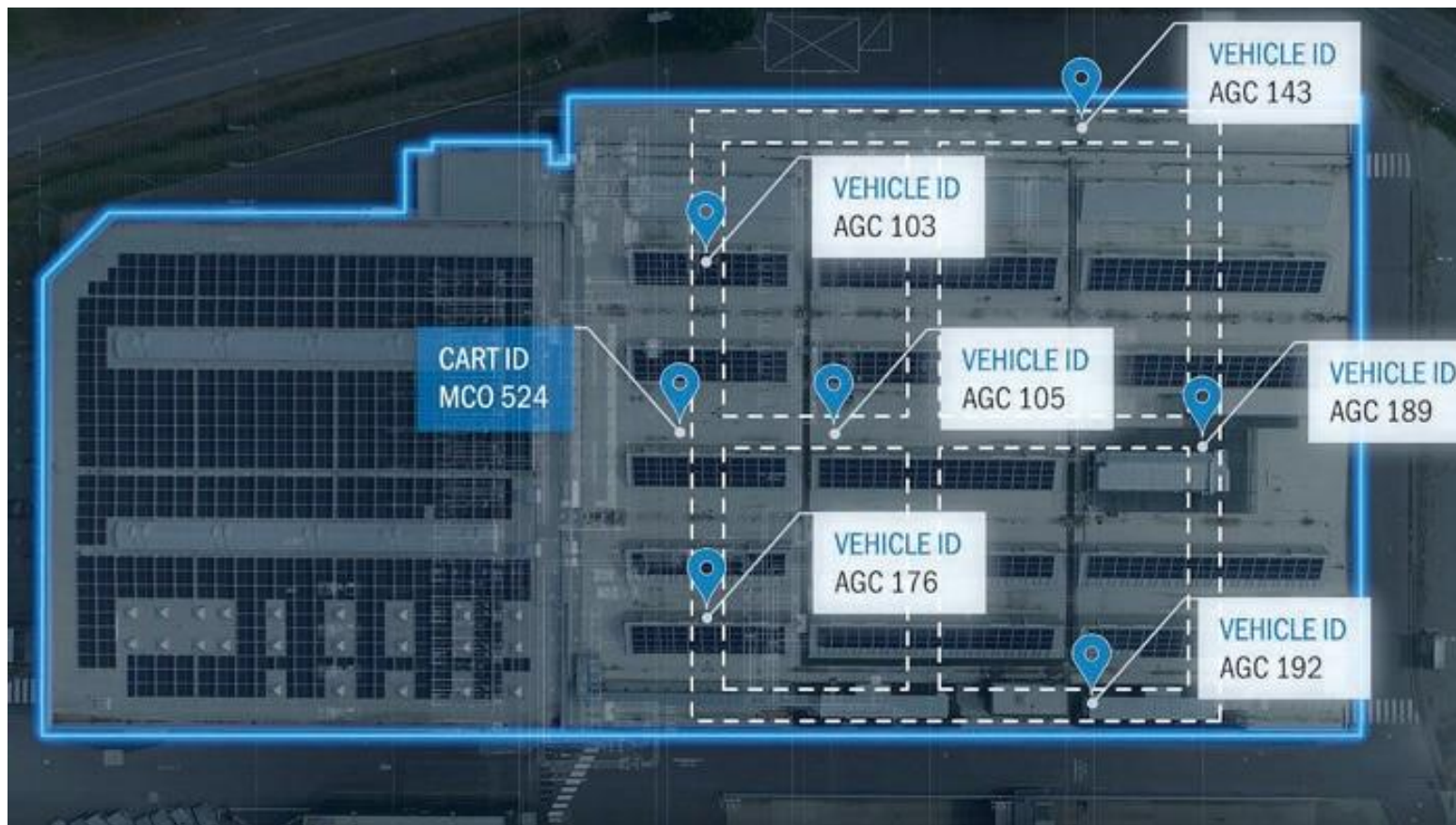


Types of Localization (Indoor Localization)





Types of Localization (Indoor Localization)





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Localization Techniques

- There are a lot of localization techniques. The most common are:
 - GPS (Global Positioning System)
 - For outdoor localization
 - WiFi-Based Localization
 - For indoor localization

